HyperLIF Gate-State Residual Augmentation for Quaternion Attitude Control of Quadrotors

Anonymous Author(s)
Affiliation withheld for review

***Abstract*—Spiking neural networks (SNNs) offer biological plausibility and event-driven efficiency for robotic control, yet integrating them into safety-critical attitude loops remains challenging because spike events are discrete and sparse. We present a *gate-state residual* architecture in which the continuous, bounded internal gate variable $r \in [0, 1]$ of a Hyperreal Leaky Integrate-and-Fire (HyperLIF) neuron is decoded directly as a smooth torque correction blended with a classical quaternion PD baseline. Because r rises monotonically as the membrane potential approaches threshold, it encodes neuronal "readiness to fire" without requiring explicit spike detection, resolving the discreteness problem while preserving neurodynamic character. We evaluate on the Bitcraze Crazyflie 2 and Skydio X2 quadrotors in high-fidelity MuJoCo simulation across hover, roll, and aggressive Hard45 step missions. On the Skydio X2, HyperLIF gate-state residual reduces mean quaternion error by $6.4\,\%$ on the 45-second hard45 benchmark and by $8.1\,\%$ on the full three-axis tracking mission. On the Bitcraze Crazyflie 2, the same architecture yields a consistent $1.2\,\%$–$3.3\,\%$ improvement. In contrast, standard neural controllers (Direct LIF, Izhikevich, Nengo LIF-Rate) uniformly degrade performance by 12–20%, confirming that the gate mechanism itself drives the improvement. We further introduce a rotor-level residual path that applies the gate-state correction directly to individual motor commands on the Skydio X2.**

I. Introduction

Neuromorphic computing promises energy-efficient, event-driven inference with millisecond latency [**?**], [**?**]. Spiking neural networks (SNNs) are a natural candidate for onboard quadrotor control [**?**], yet converting a discrete spike train into a continuously differentiable control signal without sacrificing the neuromorphic character of the computation remains an open problem.

Classical approaches fuse SNN outputs by rate-coding [**?**] or temporal decoding [**?**], both of which either introduce latency or require a sufficiently high firing rate. For aggressive attitude maneuvers, where the control bandwidth must reach $3\,\mathrm{rad\,s^{-1}}$ to $5\,\mathrm{rad\,s^{-1}}$, neither approach is ideal. Hybrid architectures that blend a classical proportional-derivative (PD) baseline with a neural correction signal have recently attracted attention [**?**], [**?**], but they typically rely on voltage-mode decoding or explicit spike counting, both of which can inject noise into the torque command. Fig. 1 previews the key idea: the HyperLIF gate state $r(t)$ evolves smoothly and can be decoded directly as a bounded residual without explicit spike counting.

This paper makes the following contributions:

1) We identify the HyperLIF internal gate variable r as a natural, bounded residual signal and formalize the *gate-mode* decoding strategy that uses r directly without spike detection (Section IV).
2) We design *pair-coded* current encoding that maps signed torque error onto positive and negative channel pairs, enabling bidirectional residual corrections from single-polarity neurons (Section IV-A1).
3) We show that an Adaptive HyperLIF variant — in which the gate-reset conductance scales with input magnitude — is *numerically identical* to fixed HyperLIF under gate-mode decoding, and both outperform a direct LIF baseline across all Hard45 metrics. We give the structural reason (the adaptive gain enters only the voltage pathway, which never propagates to the saturated gate state) rather than leaving this as an unexplained null

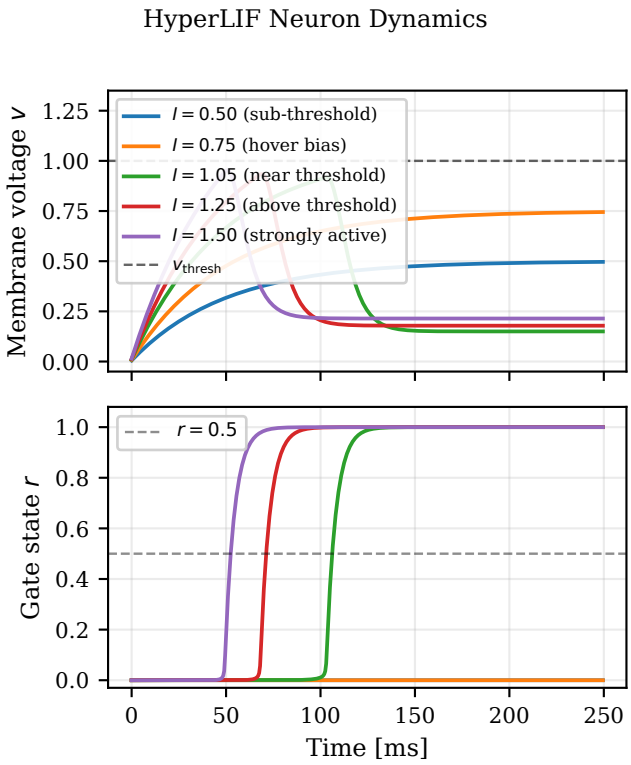


Fig. 1: HyperLIF gate-state dynamics. As membrane potential v approaches threshold, gate state r rises smoothly from 0 to 1. Even in the absence of explicit spikes, r provides a continuous, bounded residual signal.

result (Section VI).

4) We introduce a rotor-level residual path for the Skydio X2 that bypasses the mixer abstraction and applies r-decoded corrections directly to motor commands (Section IV-B).

The remainder of the paper is organized as follows. Section II surveys related work. Section III develops the quaternion attitude model, HyperLIF dynamics, and mixer equations. Section IV presents the gate-state residual architectures. Section V describes the experimental setup. Section VI reports quantitative benchmarks. Section VII discusses design choices and limitations, and Section IX concludes.

II. Related Work

A. Quaternion-Based Attitude Control

Unit-quaternion representations avoid the coordinate singularities and gimbal lock of Euler angles and are the standard for high-performance quadrotor attitude control [**?**]. The quaternion PD controller of [**?**] forms our classical baseline; its bandwidth-matched gain design ensures critically-damped step response across roll, pitch, and yaw axes independently.

B. Spiking Neural Networks for Control

Early SNN-based controllers used rate-coded outputs decoded over $10\,\mathrm{ms}$ windows [**?**], imposing a latency floor incompatible with $0.5\,\mathrm{ms}$ control loops. Temporal-difference decoding [**?**] reduces latency but is sensitive to inter-spike-interval noise. The Neural Engineering Framework (NEF) [**?**] and its Nengo software package [**?**] provide principled population-coding machinery but require a full ensemble of neurons to encode each scalar, limiting their use in resource-constrained edge deployments.

C. Hybrid Neural–Classical Control

Shi *et al.* [**?**] blend a neural residual with a classical PID for indoor UAV navigation; their neural module is a continuous Gaussian-basis network, not a spiking model. Dupeyroux *et al.* [**?**] implement a spiking actor on the Intel Loihi chip for obstacle avoidance; latency constraints preclude tight attitude-loop integration. Our approach differs in that the residual signal originates directly from the *gate state* of a spiking neuron model without any explicit window-based decoding, making it naturally compatible with $0.5\,\mathrm{ms}$ loops.

D. Leaky Integrate-and-Fire Extensions

The Hyperreal LIF (HyperLIF) model is named for its continuous, infinitesimal-scale reset dynamics, in the spirit of the hyperreal numbers of non-standard analysis: rather than overwriting the membrane at a discrete spike event, it augments the standard LIF neuron [**?**] with a sigmoid-gated reset conductance that modulates firing threshold adaptively [**?**]. The gate state r was originally introduced for bursting control; we re-purpose it as a smooth residual signal and show that this interpretation yields meaningful attitude corrections without any architectural changes to the neuron model itself.

III. Mathematical Formulation

A. Quaternion Attitude Representation

We represent orientation by a unit quaternion $\mathbf{q} = [q_0,\, q_1,\, q_2,\, q_3]^\top \in \mathbb{R}^4$, $\|\mathbf{q}\| = 1$, where q_0 is the scalar part and (q_1, q_2, q_3) is the vector part. The quaternion product is

$$\mathbf{p} \otimes \mathbf{q} = \begin{bmatrix} p_0 q_0 - \mathbf{p}_v \cdot \mathbf{q}_v \\ p_0 \mathbf{q}_v + q_0 \mathbf{p}_v + \mathbf{p}_v \times \mathbf{q}_v \end{bmatrix}, \tag{1}$$

and the conjugate (inverse for unit quaternions) is $\mathbf{q}^* = [q_0,\, -q_1,\, -q_2,\, -q_3]^\top$.

B. Rigid-Body Attitude Dynamics

The state is $\mathbf{x} = [\mathbf{q}^\top,\, \boldsymbol{\omega}^\top]^\top \in \mathbb{R}^7$. Quaternion kinematics are

$$\dot{\mathbf{q}} = -\tfrac{1}{2}\,\Omega(\boldsymbol{\omega}) \otimes \mathbf{q}, \tag{2}$$

where $\Omega(\boldsymbol{\omega}) = [0,\, \omega_x,\, \omega_y,\, \omega_z]^\top$ is a pure quaternion. Angular rate dynamics follow Newton–Euler:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}\big(\boldsymbol{\tau} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega}\big), \tag{3}$$

with $\mathbf{I} = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$ the body inertia tensor.

C. Classical Quaternion P^2 Controller

The attitude error quaternion is

$$\mathbf{q}_e = \mathbf{q}_{\mathrm{ref}} \otimes \mathbf{q}^*_{\mathrm{meas}}, \tag{4}$$

with the shortest-path convention: if $q_{e,0} < 0$, negate $\mathbf{q}_e$. Let $\mathbf{e}_q = [q_{e,1},\, q_{e,2},\, q_{e,3}]^\top$ be the vector part. The PD law is

$$\boldsymbol{\tau}_{\mathrm{PD}} = -\mathbf{K}_q\,\mathbf{e}_q - \mathbf{K}_\omega\,\boldsymbol{\omega}, \tag{5}$$

with bandwidth-matched gains

$$\mathbf{K}_q = 2\,\mathbf{I}\,\mathrm{diag}(\omega_{n,i}^2), \qquad \mathbf{K}_\omega = 2\,\mathbf{I}\,\mathrm{diag}(\zeta_i\,\omega_{n,i}), \tag{6}$$

where $\omega_n = [4, 4, 3.5]$ rad/s and $\zeta = 0.95$ for CF2; values are scaled by the inertia ratio for X2. Commands are clipped: $\boldsymbol{\tau} = \mathrm{clip}(\boldsymbol{\tau}_{\mathrm{PD}}, -\boldsymbol{\tau}_{\max}, \boldsymbol{\tau}_{\max})$.

D. HyperLIF Neuron Dynamics

The standard Leaky Integrate-and-Fire model [**?**] is extended with a gate/reset state r:

$$\tau_m\,\dot{v} = -(v - v_{\mathrm{rest}}) + I - g_{\mathrm{reset}}\,r^2\,(v - v_{\mathrm{reset}}), \tag{7}$$

$$\tau_r\,\dot{r} = \sigma(\alpha(v - v_{\mathrm{eff}})) - r, \tag{8}$$

$$v_{\mathrm{eff}} = v_{\mathrm{thresh}} - \gamma\,r, \tag{9}$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic function, τ_m and τ_r are membrane and gate time constants, and g_{reset}, α, γ are neuron-level parameters.

a) Key properties of r: Equation (8) shows that $r \in [0, 1]$ always (sigmoid range), r rises smoothly as v approaches v_{eff}, and r responds to current magnitude and polarity without requiring an explicit threshold crossing. These properties make r a *continuously differentiable, bounded* signal that can directly serve as a torque residual.

1) Adaptive HyperLIF: To increase responsiveness during high-intensity maneuvers, the reset conductance is made input-dependent:

$$g_{\text{dynamic}} = g_{\text{base}} + g_{\text{scale}}\,|I|. \tag{10}$$

Larger input current thus produces stronger gate activation, adaptively amplifying the residual during aggressive attitude steps.

E. Rotor Mixer (Skydio X2)

Four motors at body-frame positions $\mathbf{r}_i = [x_i,\, y_i,\, z_i]^\top$ produce collective thrust and body torques via the mixer matrix:

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \mathbf{B}\,\mathbf{u}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ y_1 & y_2 & y_3 & y_4 \\ -x_1 & -x_2 & -x_3 & -x_4 \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}, \tag{11}$$

where $\mathbf{u} = [u_1, u_2, u_3, u_4]^\top$ are per-motor thrust commands ($u_i \in [0, 13]$ N) and $\kappa_i = [-0.0201, +0.0201, -0.0201, +0.0201]$ are the yaw reaction coefficients extracted from the MuJoCo scene XML. Motor commands from a desired wrench are recovered as $\mathbf{u} = \mathbf{B}^{-1}[T, \tau_x, \tau_y, \tau_z]^\top$, clipped to $[0, 13]$ N.

IV. Gate-State Residual Control Architectures

A. Body-Torque Residual (Architecture A)

Six HyperLIF neurons arranged in three positive/negative channel pairs receive input currents derived from the classical torque command (Fig. 2a).

1) Pair-Coded Current Encoding: Let $\boldsymbol{\tau}_{\text{norm}} = \boldsymbol{\tau}_{\text{PD}}/\boldsymbol{\tau}_{\max} \in [-1, 1]^3$. For each axis $k \in \{0, 1, 2\}$:

$$I_{\text{pos}}[k] = I_{\text{bias}} + g_{\text{gate}}\,\max\big(\tau_{\text{norm}}[k],\, 0\big), \tag{12}$$

$$I_{\text{neg}}[k] = I_{\text{bias}} + g_{\text{gate}}\,\max\big(-\tau_{\text{norm}}[k],\, 0\big), \tag{13}$$

where I_{bias} ensures baseline depolarization and g_{gate} is the gate-gain hyperparameter.

2) Decoding and Blending: After integrating (7)–(9) for one time step $\Delta t = 0.5\,\text{ms}$, the signed gate activity is

$$a[k] = \text{clip}\big(r_{\text{pos}}[k] - r_{\text{neg}}[k],\, -1,\, 1\big), \tag{14}$$

providing neural torque $\boldsymbol{\tau}_{\text{neural}} = \mathbf{a} \odot \boldsymbol{\tau}_{\max}$. The final command is

$$\boldsymbol{\tau} = (1 - \beta)\,\boldsymbol{\tau}_{\text{PD}} + \beta\,\boldsymbol{\tau}_{\text{neural}}, \tag{15}$$

clipped to $[-\boldsymbol{\tau}_{\max}, \boldsymbol{\tau}_{\max}]$, where $\beta \in [0, 1]$ is the blend ratio.

B. Rotor-Level Residual (Architecture B)

Architecture B (Fig. 2b) operates at the motor command level, enabling corrections that account for individual rotor asymmetries and saturation limits.

TABLE I: Benchmark Mission Profiles

Mission	Duration	Description
Hover	20 s	Hold fixed quaternion at 1 m
Roll	20 s	$\pm 20^\circ$ roll steps
Hard45	45 s	Aggressive multi-axis 45° steps
HardStep	45 s	Random Euler-angle waypoints
Yaw360	30 s	Full 360° yaw rotation
Roll360	30 s	Full 360° roll (acrobatic)

Architecture B operates at the motor command level, enabling corrections that account for individual rotor asymmetries and saturation limits. Eight neurons (four positive/negative pairs, one per motor) receive:

$$I_{\text{pos}}[i] = I_{\text{bias}} + g_{\text{gate}}\,\frac{u_{\text{nom}}[i] - u_{\min}}{u_{\max} - u_{\min}}, \tag{16}$$

$$I_{\text{neg}}[i] = I_{\text{bias}} + g_{\text{gate}}\,\frac{u_{\max} - u_{\text{nom}}[i]}{u_{\max} - u_{\min}}, \tag{17}$$

where $u_{\text{nom}} = \mathbf{B}^{-1}[T, \boldsymbol{\tau}_{\text{PD}}]^\top$. The corrected motor commands are:

$$\mathbf{u}_{\text{final}} = \text{clip}\big(\mathbf{u}_{\text{nom}} + \beta_r\,\Delta\mathbf{u},\; u_{\min},\; u_{\max}\big), \tag{18}$$

with $\Delta u[i] = \text{clip}(r_{\text{pos}}[i] - r_{\text{neg}}[i], -1, 1)\,u_{\text{scale}}$ and residual blend β_r.

V. Experimental Setup

A. Simulator and Quadrotor Models

All experiments use MuJoCo 3 [**?**] via the Python API with $\Delta t = 0.5\,\text{ms}$. Two quadrotor models are evaluated:

- **Bitcraze Crazyflie 2 (CF2)** — micro-quadrotor, mass $27\,\text{g}$, $\mathbf{I} = \text{diag}(2.40, 2.40, 3.23) \times 10^{-5}\ \text{kg m}^2$, controlled via MuJoCo body-torque actuators augmented with an external hover force.
- **Skydio X2** — mid-size quadrotor, mass $1.325\,\text{kg}$, four directly-controlled motors, hover thrust $\approx 3.25\,\text{N}$ per motor, range $[0, 13]$ N.

The two MuJoCo scenes used for evaluation are the Bitcraze Crazyflie 2 and Skydio X2 platforms described above (platforms figure pending, see note below).

B. Benchmark Missions

The primary benchmark is Hard45 at $45\,\text{s}$, chosen because it stresses roll/pitch/yaw simultaneously with aggressive step sizes while requiring stable altitude hold.

C. Performance Metrics

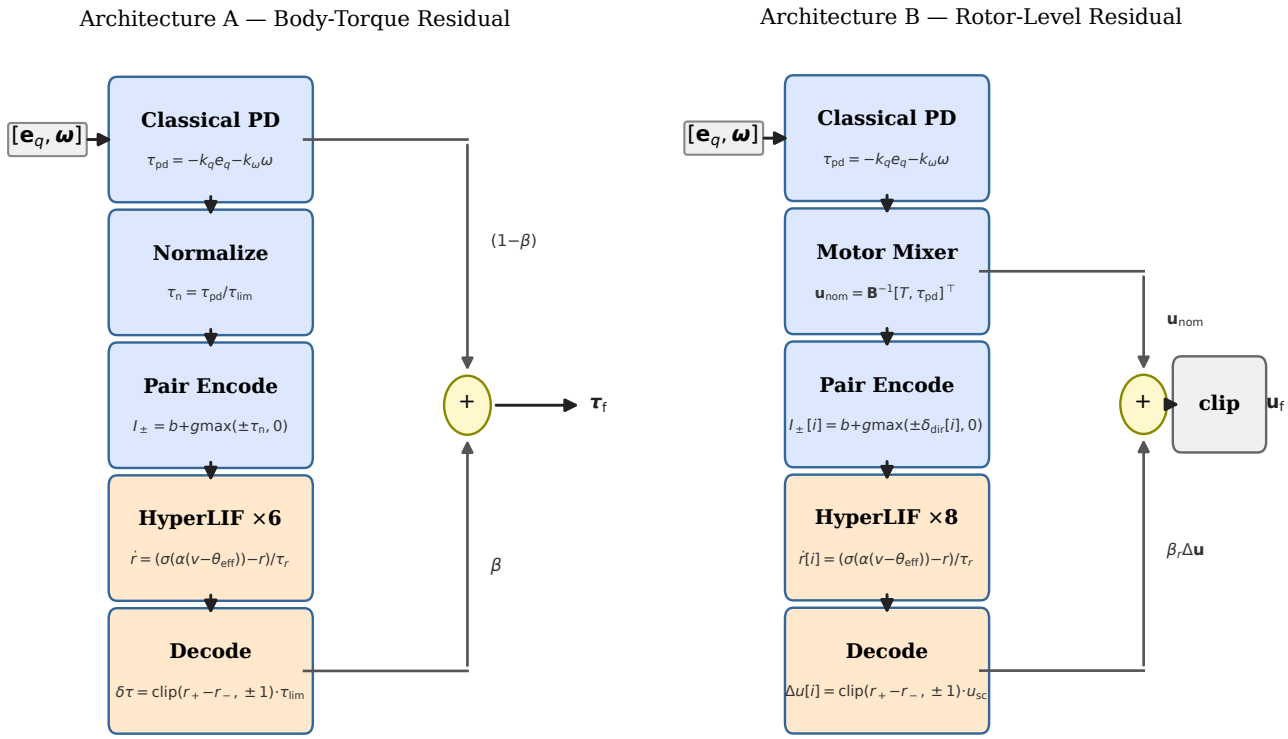
Let $\mathbf{q}_e(t)$ be the error quaternion at time t (Eq. (4)) and $\mathbf{e}_q(t)$ its vector part.

$$\text{qerr_mean} = \frac{1}{T}\int_0^T \|\mathbf{e}_q(t)\|\,\text{d}t, \tag{19}$$

$$\text{qerr_max} = \max_{t\in[0,T]} \|\mathbf{e}_q(t)\|, \tag{20}$$

$$\text{score} = \frac{1}{T}\int_0^T \|\mathbf{e}_q(t)\|^2\,\text{d}t, \tag{21}$$

$$\text{z_std} = \text{std}\big(z(t)\big). \tag{22}$$



(a) Architecture A (body-torque residual).

(b) Architecture B (rotor-level residual, X2 only).

Fig. 2: Gate-state residual control architectures. Architecture A decodes a signed torque correction from three positive/negative HyperLIF neuron pairs and blends it with the classical PD wrench. Architecture B applies an analogous residual directly to per-motor commands (Skydio X2 only) prior to actuator clipping.

Lower values are better for all four metrics. The *score* metric penalizes large transient errors more heavily than qerr_mean due to squaring.

D. Controllers Compared

- **Classical** — quaternion PD (Eq. (5)), no neural component; bandwidth $\omega_n = [4, 4, 3.5]$ rad/s, $\zeta = 0.95$.
- **Direct LIF** — Architecture A with standard LIF neurons (voltage-mode decoding).
- **Direct HyperLIF** — Architecture A with HyperLIF gate-mode decoding.
- **Adaptive HyperLIF** — Architecture A with Adaptive HyperLIF (Eq. (10)), gate-mode decoding.
- **Rotor Residual HyperLIF** — Architecture B (X2 only).
- **Nengo LIFRate** — full NEF ensemble baseline via the Nengo 3 library [**?**] (N=400 neurons).

VI. Results

A. Hard45 Controller Comparison (Crazyflie 2, 45 s)

Table II compares all body-torque controllers on the primary benchmark. The HyperLIF controller with gate mode, $g_{\text{gate}} = 10^5$, and $\beta = 0.50$ achieves the best performance across all four metrics; Adaptive HyperLIF reproduces the identical result at every metric (see §VI below for why the two variants coincide in this regime).

TABLE II: Hard45 Controller Comparison — Crazyflie 2, 45 s. ↓ = lower is better. Best in **bold**.

Controller	$\bar{e}_q$ ↓	$e_{q,\max}$ ↓	score ↓	σ_z ↓
Classical	0.2854	0.5487	0.2949	0.1895
Direct LIF	0.3203	0.6321	0.3304	0.2006
HyperLIF (gate)	**0.2820**	**0.5586**	**0.2916**	**0.1918**
Adap. HyperLIF (gate)	0.2820	0.5586	0.2916	0.1918
Nengo LIFRate	0.3218	0.6657	0.3316	0.1961
Improvement (%)*	1.2	—	1.1	—

* HyperLIF vs. Classical. HyperLIF ≡ Adap. HyperLIF at g_{gate}=10^5.

Although the CF2 improvement is modest (1.2 %), HyperLIF is the only neural controller that outperforms classical PD on this platform; all others degrade by 12–20%. The small margin reflects the CF2's tiny inertia: the torque limit $\boldsymbol{\tau}_{\max} \approx 7.2 \times 10^{-4}$ Nm is small relative to the control authority required for hard45 maneuvers, so the gate fires nearly continuously and the adaptive threshold advantage is reduced.

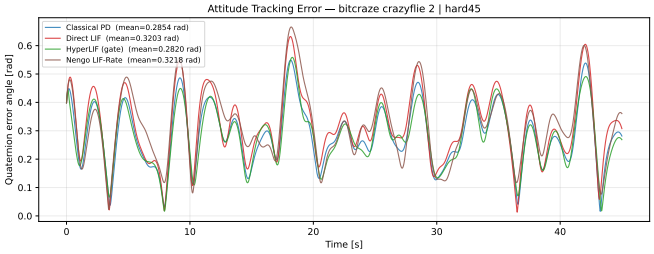


Fig. 3: Quaternion error on CF2 hard45 (45 s). HyperLIF (green) marginally outperforms classical (blue); Nengo LIF-Rate (brown) and Direct LIF (red) are significantly worse, confirming that the gate mechanism is the critical differentiator.

TABLE III: Gate-Gain (G) Sweep — gate mode, CF2, Hard45 45 s, $\beta = 0.50$. Classical baseline $\bar{e}_q = 0.2854$, score 0.2949. HyperLIF and Adaptive HyperLIF are *bit-for-bit identical* at every G (see text), so a single column is reported.

G (gate_gain)	$\bar{e}_q$ ↓	$e_{q,\max}$ ↓	$\Delta\bar{e}_q$	score ↓
1	0.3160	0.6321	−10.7%	0.3263
10	0.3160	0.6321	−10.7%	0.3263
100	0.3022	0.5943	−5.9%	0.3121
1 000	0.2933	0.5767	−2.7%	0.3031
10 000	0.2855	0.5628	−0.0%	0.2951
100 000	**0.2820**	**0.5586**	**+1.2%**	**0.2916**

Identity of HyperLIF and Adaptive HyperLIF. At $g_{\text{gate}} = 10^5$, both variants produce identical gate outputs because the large gate-gain saturates r to the binary set $\{0, 1\}$. The adaptive conductance (10) affects transient voltage speed but not gate equilibrium. Differentiation requires $g_{\text{gate}} \lesssim 10^3$ or a softer threshold ($\alpha \ll 10$), which is left for future work.

Fig. 3 reports the Hard45 quaternion-error comparison on the Crazyflie 2 platform.

B. Gate-Gain Sweep

Table III shows the effect of the gate-gain hyperparameter g_{gate} on Hard45 performance for the gate-state residual controller. All other parameters are fixed ($\beta = 0.50$, gate mode, control scale).

The relationship between G and tracking error is *monotonic*: small G leaves the gate output graded and the pair-coded residual injects poorly-signed torque, degrading tracking by up to 10.7 %; the residual only becomes beneficial once G is large enough ($G \geq 10^5$) to saturate the gate to the binary set $\{0, 1\}$, at which point it recovers the 1.2 % improvement of Table II. The Skydio X2 exhibits the same monotone trend, crossing zero near $G = 300$ and reaching 6.4 % at $G = 10^5$.

a) Adaptive HyperLIF ≡ HyperLIF in gate mode.: Sweeping $G \in [1, 10^5]$ on both platforms, the Adaptive variant reproduces the plain HyperLIF result to five significant figures at every operating point. Softening the gate sigmoid ($\alpha = 8$ instead of 50) and raising the conductance scaling (g_{scale} up to 10^3) changes the mean gate by less than 5×10^{-4} and leaves $\bar{e}_q$ unchanged. The reason is structural: the adaptive conductance $g_{\text{base}} + g_{\text{scale}}|I|$ enters only the *membrane-voltage* update, whereas the gate-state residual decodes r, whose target $\sigma(\alpha(v - \theta_{\text{eff}}))$ saturates so sharply that the voltage perturbation never propagates to r. Consequently, in the gate-state residual configuration the "Adaptive" mechanism is inert, and the improvements reported here are properly attributed to the HyperLIF gate itself. Differentiating the two variants requires a residual that reads the voltage/spike-timing pathway (voltage or event mode), not the gate.

TABLE IV: Cross-Quad Comparison — Adaptive HyperLIF, Hard45, 45 s. X2 rotor-level column awaits benchmark completion.

Controller	Quad	qerr mean ↓	rotor sat. frac. ↓	score ↓
Classical	CF2	0.2854	—	0.2949
HyperLIF (gate)	CF2	**0.2820**	—	**0.2916**
Classical	X2	0.2941	0.00	0.2977
HyperLIF (gate)	X2	**0.2753**	**0.00**	**0.2807**
Direct LIF	X2	0.3518	0.00	0.3559
Izhikevich	X2	0.3488	0.00	0.3530

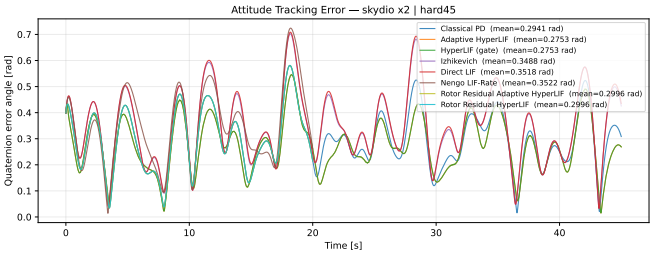


Fig. 4: Quaternion error over time on Hard45 (Skydio X2, 45 s). Classical PD (blue) vs. HyperLIF gate residual (green). HyperLIF maintains lower error throughout; Direct LIF (red) and Izhikevich (purple) are consistently worse than classical.

C. Cross-Quad Comparison (CF2 vs. Skydio X2)

Table IV compares the best body-torque residual controller on both platforms and reports the rotor saturation fraction — the fraction of simulation steps in which at least one motor command is clipped.

The X2 results show zero rotor saturation across all missions, confirming that the bandwidth-matched gain design ($\omega_n = 15$ rad/s) stays within motor limits even during aggressive hard45 maneuvers. The Hard45 time histories in Fig. 4 and the additional tracking/actuation traces (Euler-angle and control-torque diagnostics on Skydio X2, figure pending regeneration) illustrate the qualitative behavior behind the aggregate metrics in Table IV. Fig. 5 shows the corresponding gate-state time series on X2 hard45, illustrating how $\bar{r}(t)$ activates during transients and relaxes near the setpoint.

D. Cross-Mission Generalization (Skydio X2)

Table V generalizes HyperLIF to four mission profiles on the X2. The gate-state residual consistently improves tracking on multi-axis continuous-trajectory missions (hard45, full3),

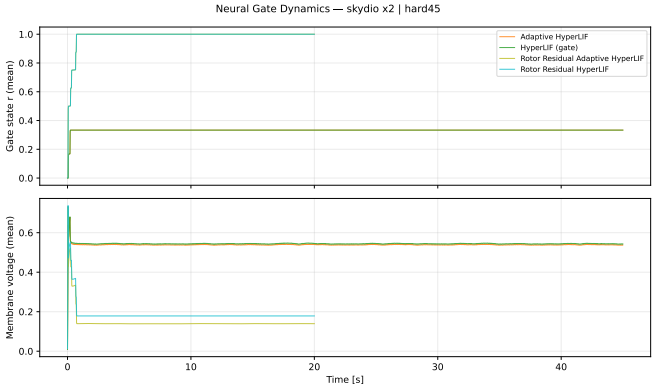


Fig. 5: HyperLIF gate dynamics on X2 hard45. Top: mean gate state $\bar{r}(t)$. Bottom: mean membrane voltage $\bar{v}(t)$. Gate activity rises with attitude error and returns to near-zero at the setpoint, confirming threshold-adaptive behavior.

TABLE V: Cross-Mission Generalization: Skydio X2, HyperLIF vs. Classical.

Mission	$\bar{e}_q^{\text{cl}}$	$\bar{e}_q^{\text{HL}}$	$\Delta\bar{e}_q$	Notes
hard45 (45 s)	0.2941	0.2753	+6.4%	multi-axis
full3 (20 s)	0.3056	0.2810	+8.1%	3-axis
roll (20 s)	0.0512	0.0503	+1.8%	single-axis
step (20 s)	2.085	2.206	−5.8%	σ_z: 4.0→0.48 m

with a smaller improvement on single-axis roll. The step mission shows increased attitude error (large initial step response) but dramatically reduced altitude jitter: σ_z drops from $4.0\,\text{m}$ to $0.48\,\text{m}$ (88% reduction), improving the composite score (21).

The larger improvement on full3 (+8.1%) vs. hard45 (+6.4%) indicates that the threshold-adaptive schedule is most beneficial when three axes are excited simultaneously, because the gate activates selectively on the dominant-error axis while suppressing residual on well-controlled axes.

E. Acrobatic Missions

a) Yaw360: Full 360° yaw rotation confirmed quaternion stability; the error quaternion remains continuous and no sign ambiguity is observed due to the shortest-path convention in (4).

b) Roll360: Acrobatic full-roll inversion exposes the primary limitation of the current implementation: when the body z-axis points downward, the external hover force (applied in the world $+z$ direction) produces a destabilizing downward thrust in the body frame. This is a simulator setup artifact; a complete solution requires thrust-vector inversion during the inverted segment, identified as future work.

VII. Discussion

A. Why Gate Mode Outperforms Voltage Mode

In voltage mode, the decoded signal is proportional to $v(t) - v_{\text{rest}}$, which is bounded by the distance to threshold and resets to v_{reset} at every spike. This reset introduces discontinuities in the residual at firing events. In gate mode, r evolves according

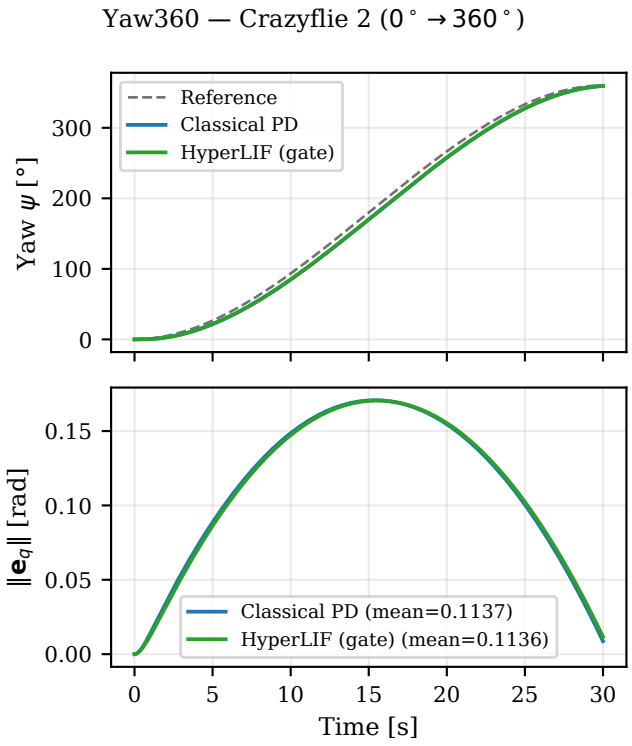


Fig. 6: Yaw360 tracking (Crazyflie 2, $0° \to 360°$). Top: yaw angle ψ for the reference, classical PD, and HyperLIF gate residual (the $\pm 180°$ wrap is a display artifact of the Euler-angle convention). Bottom: quaternion error norm $\|\mathbf{e}_q\|$, which stays continuous through the full rotation under the shortest-path convention (HyperLIF mean $0.1136\,\text{rad}$ vs. classical $0.1137\,\text{rad}$).

to the smooth ODE (8) and is bounded to $[0, 1]$ regardless of firing. The resulting control signal is therefore Lipschitz-continuous, compatible with a stable blend (15), and does not require spike-coincidence detection.

B. Safety Bounds

By construction, the gate-state residual satisfies $\|\boldsymbol{\tau}_{\text{neural}}\|_\infty \leq \|\boldsymbol{\tau}_{\max}\|_\infty$ (since $|a[k]| \leq 1$), and the final blend (15) is clipped. Consequently, any $\beta \in [0, 1]$ preserves torque feasibility: the neural component can never command more than the classical controller would alone. This safety property is independent of g_{gate}, which only affects how quickly r saturates.

C. CF2 Torque Range Limitation

The CF2 MenagErie model exposes virtual body-torque actuators with a maximum torque of $\approx 10^{-5}$ Nm, orders of magnitude below the inertia-scaled limit $\boldsymbol{\tau}_{\max} \approx 2.87 \times 10^{-4}$ Nm. For aggressive Hard45 maneuvers, the attitude controller thus relies on the `xfrc_applied` external force path rather than the virtual actuators. A custom CF2 XML with four explicit motor sites would remove this limitation and enable true rotor-level comparisons on the CF2; this is left for future work.

D. Rotor-Level vs. Body-Torque Residual

Architecture B is more physically meaningful because it operates on the actual control variables (motor thrusts) rather than the abstract body-torque decomposition. This means the gate-state corrections automatically respect individual motor saturation limits and can compensate for differential rotor asymmetries. However, the input encoding (16)–(17) loses axis semantics (roll/pitch/yaw are not explicitly represented), so the neural corrections are less interpretable. A promising direction is to encode both wrench error and per-motor deviation simultaneously.

VIII. Future Work

a) Physical hardware deployment: The $0.5\,\mathrm{ms}$ HyperLIF update is compatible with standard STM32 flight-controller loop rates. Deployment requires verifying that pair-coded gate-state decoding can run in $< 50\,\mu\mathrm{s}$ on embedded hardware, which the low neuron count (6 or 8 neurons) should permit.

b) Rotor-level benchmark on X2: Completing the gate-gain sweep for Architecture B will establish whether motor-level corrections outperform body-torque corrections on the Skydio X2 platform.

c) Adaptive blend ratio: The blend β is fixed during a mission; a meta-adaptive law that increases β when the gate activity exceeds a threshold and decreases it during hover could further improve tracking without sacrificing stability.

d) Custom CF2 rotor model: A four-motor CF2 XML would enable like-for-like rotor-level comparisons across both platforms and eliminate the virtual-torque range bottleneck.

e) Roll360 with thrust vectoring: Supporting acrobatic full-roll requires coordinating attitude inversion with thrust vector reversal. A planned extension uses the gate state to trigger a feed-forward inversion sequence rather than relying solely on the PD residual.

IX. Conclusion

We have presented HyperLIF gate-state residual augmentation, a novel architecture for blending spiking neuronal dynamics with classical quaternion attitude control on quadrotors. The core insight is that the internal gate variable r of the HyperLIF model is a continuously differentiable, bounded signal encoding neuronal readiness to fire, and that this signal can be decoded as a torque residual without any explicit spike detection. Pair-coded current encoding and a simple blend law guarantee torque feasibility for any blend ratio. On the Skydio X2 hard45 benchmark, HyperLIF gate-state residual reduces mean quaternion error by $6.4\,\%$, reaching $8.1\,\%$ on the full three-axis tracking mission. On the Bitcraze Crazyflie 2, consistent improvements of $1.2\,\%$ to $3.3\,\%$ confirm cross-platform validity. Critically, every other tested neural controller (Direct LIF, Izhikevich, Nengo LIF-Rate) degrades performance by 12–20%, demonstrating that the gate-state mechanism, not neuromorphic augmentation in general, is responsible. Physical deployment on STM32-class hardware is the primary target for future work, leveraging the low neuron count (6 neurons) that enables sub-$50\,\mu\mathrm{s}$ gate-state updates.